Supplementary Material: Explicit Frictional Contact Modeling in Neural Garment Simulation

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S1. Operator implementation

We apply the operator at inference to a frozen backbone, preserving predictor weights, timestep, and material parameters. Table S1 summarizes the settings used in all reported evaluations. All non-input settings are fixed across sequences and friction coefficients.

Activation. The proximity set $\mathcal{C}_\varepsilon$ selects vertices for inspection. However, proximity alone does not activate friction. Instead, a vertex receives a tangential correction only when the positive bracket in Eq. (4) yields a positive estimated load. When $\widehat{\lambda}_n = 0$, the arrest length is zero and the predicted tangential displacement remains unchanged. The final non-penetration step is a geometric repair and does not change this activation decision.

Triangle walk and completion. The traversal of Sec. 4.4 retains each endpoint's signed supporting-plane gap and walks its in-plane foot across adjacent faces by project–clip–unfold steps. At a crossed edge, a rigid hinge rotation develops the target into the next face and the lifted junction follows the bisector of the adjacent face normals; the development is isometric over the traversed face strip, although it is not claimed to be a shortest geodesic. The arrest length is re-evaluated on each face, and the walk selects the closest admissible landing to the tentative endpoint under Eq. (7). Same-face pairs use the exact segment; a boundary, degenerate face, ambiguous normal, or exhausted crossing budget reverts that vertex to the straight-chord landing.

After traversal, a fixed local-plane refinement restores the proposal's clearance, subject to the non-penetration slack in Table S1. Let $\boldsymbol{\delta}_{T,i}^{\mathrm{fric}} = \boldsymbol{\Delta}_{T,i}^{+} - \boldsymbol{\Delta}_{T,i}$ be the tangential correction and let $\boldsymbol{\delta}_i^{\mathrm{comp}}$ be the accumulated completion displacement. For an unambiguous correspondence, completion is accepted only when $\|\boldsymbol{\delta}_i^{\mathrm{comp}}\| \le c_i$, with $c_i = \max(2\varepsilon, \|\boldsymbol{\delta}_{T,i}^{\mathrm{fric}}\|)$; an ambiguous completion or one exceeding this scale retains the tangential landing. This per-vertex rule is a revert safeguard, not a displacement clip.

S1.1. Propagation numerics

We integrate Eq. (8) on the free vertices with symplectic Euler,

$$\dot{\mathbf{w}} \leftarrow \dot{\mathbf{w}} - \Delta\tau\, \mathbf{M}^{-1}\mathbf{K}_0\mathbf{w}, \qquad \mathbf{w} \leftarrow \mathbf{w} + \Delta\tau\, \dot{\mathbf{w}}, \quad \text{(S1)}$$

where $\mathbf{K}_0$ is linearized at the tentative state and held fixed for the frame. No system is assembled or inverted. With $\lambda_{\max}$ a per-frame Gershgorin bound on the largest eigenvalue of $\mathbf{M}^{-1}\mathbf{K}_0$, we set

$$\Delta\tau_{\max} = 2/\sqrt{\lambda_{\max}}, \qquad n_{\mathrm{sub}} = \lceil h/(0.8\,\Delta\tau_{\max}) \rceil \quad \text{(S2)}$$

and round upward on the ladder in Table S1. If the required count exceeds its largest rung, propagation returns the pre-propagation result rather than integrating outside the bound. The stage has no damping and retains no state between frames.

All corrected contacts are driven simultaneously by $\mathbf{w}_i(\tau) = s(\tau/h)\mathbf{c}_{T,i}$, where $s(r) = 3r^2 - 2r^3$ starts and ends at rest; other contacts and pinned vertices are zero boundaries. This is the unique cubic satisfying $s(0) = 0$, $s(1) = 1$, and $s'(0) = s'(1) = 0$, so it reaches the prescribed correction without introducing a ramp parameter. Because the propagated field has not passed through the predictor's material solve, composing it can overstretch local edges. We therefore follow established cloth strain-limiting practice [1, 3, 8]: after composition, we apply 24 parallel edge-relaxation passes to the added field $\mathbf{w} = \mathbf{x}_{\mathrm{out}} - \hat{\mathbf{x}}^{t+1}$. In each pass, violating edges propose inverse-mass corrections [2] that are averaged at shared vertices, while contacts and pins remain fixed. A final projection restores any body gap reduced at a moved free vertex. Algorithm S1 summarizes the implemented order.

Project page: https://trthanhnguyen.github.io/explicit-friction-ngs/

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Table S1. Settings that define the reported operator. *Inherited* values follow the backbone or prior contact model; *derived* values are recomputed from the current frame; *selected* values were fixed before evaluation; *safeguards* are termination guards.

Setting	Value	Class	Origin
Proximity shell ε	3 mm	inherited	Backbone collision tolerance plus 1 mm.
Static band ϵ_v	$2 \times 10^{-3}\,\mathrm{m\,s^{-1}}$	inherited	The static–sliding regularizer of barrier-based contact [6].
Normal completion	0.1 mm slack	selected	Fixed local clearance refinement; unsafe correspondences revert.
Triangle-walk crossing cap	16	safeguard	Termination guard; unused in the evaluated rollouts.
Propagation substeps	0.8 margin; ladder 128–2048	derived	Framewise spectral stability bound.
Added-field strain limit	$\gamma_{\max} = 0.2$; 24 passes	selected	Per-edge budget on the transported field; provenance below.
Timestep h	$1/30$ s	inherited	Neural-backbone timestep.
Friction coefficient μ	supplied at inference	input	User-controlled operator input.

Algorithm S1 Friction-Induced Motion Propagation for one frame.

Require: tentative state $\hat{\mathbf{x}}^{t+1}$; corrected contact state $\mathbf{x}^c$; pins $\mathcal{P}$; h; $\gamma_{\max}$
$\mathcal{A} \leftarrow$ contacts with a nonzero tangential correction; $\mathcal{F} \leftarrow$ non-contact, unpinned vertices
if $\mathcal{A} = \emptyset$ **then**
  return $\mathbf{x}^c$
end if
linearize $\mathbf{K}_0$ at $\hat{\mathbf{x}}^{t+1}$ and derive n_{sub} from Eq. (S2)
if the substep ladder is exhausted **then**
  return $\mathbf{x}^c$
end if
initialize $\mathbf{w}, \dot{\mathbf{w}} \leftarrow \mathbf{0}$
for $k = 1, \ldots, n_{\text{sub}}$ **do**
  advance $\mathcal{F}$ by Eq. (S1)
  impose the contact ramp on $\mathcal{A}$
end for
set free vertices to $\hat{\mathbf{x}}^{t+1} + \mathbf{w}$
retain $\mathbf{x}^c$ at contacts and pins
apply 24 parallel edge-relaxation passes to $\mathcal{F}$
restore reduced body gaps once
return the composed state

Before the reported evaluation, calibration showed two scales in the added field: propagating the Coulomb corrections induced around 15–25 % edge strain, while source-adjacent outliers reached 100–500 % on roughly 1 % of edges. We fixed $\gamma_{\max} = 0.2$ at the former scale, limiting its upper end while clipping the outlier tail. The limiter acts only on propagation's added displacement rather than re-projecting the predictor's original deformation. Calibration outcomes were stable over 0.05–0.4, placing 0.2 near the middle of the measured plateau. Figure S3 combines these calibration scales with the full-grid sensitivity in Sec. S4.1.

S2. Normal-load validation

The estimate of Sec. 4.2 is a central physical approximation of the operator. We isolate its gravitational and impact terms on prescribed scenes with closed-form per-vertex loads, reading each estimate through the production contact path at $\mu = 0$. We report $q = \lambda_n/(mg)$ and compare with gravity-only, impact-only, and calibrated barrier-gradient controls. ARGUS [5] is used for these diagnostics because it reports nodal contact impulses directly; PPF remains the rollout reference in the paper.

Gravitational response. A 97×21 vertex cap rests at a uniform 2 mm gap over a cylinder of radius 0.15 m and is read on its first 10 ms step from rest. The analytic load is $m_i g[\cos\phi_i]_+$. Our estimate follows this profile to numerical precision (normalized RMSE 4×10^{-6} % over interior vertices), while ARGUS has normalized RMSE 1.7 % and changes by less than 3 % under timestep and mesh refinement. The calibrated barrier gradient remains near $q = 1$ at every angle (normalized RMSE 47 %) because a fixed gap does not encode surface orientation; gravity only coincides with our estimate and impact only is zero.

Impact response. A 21×21 vertex patch is dropped from rest and arrested after K flight steps. The closed-form step response is $q^* = [1 + a_n/g]_+$: for $K = 10$, it rises from zero in free fall to 11 on the arrest step. Our estimate reproduces that profile to numerical precision and ARGUS reaches the scheduled-step peak within 0.1 %; gravity only and the barrier gradient remain at 1, while impact only is short by one weight. The same pattern holds for $K \in \{1, 2, 5, 10, 20\}$, corresponding to 1–206 mm drops, and varying timestep and resolution changes the normalized profile by less than 0.5 %.

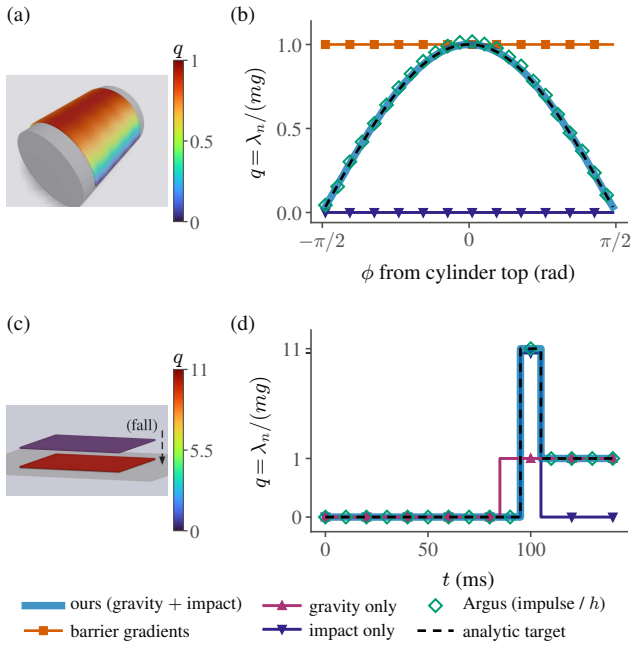


Figure S1. **Analytic probes of the normal-load estimate.** Colour and ordinates show $q = \lambda_n/(mg)$. **(a,b)** A strain-free cloth cap on a cylinder tests the gravitational profile against surface angle ϕ. **(c,d)** A flat patch dropped 54 mm onto a plane tests the impact response at release and arrest ($K = 10$, $h = 10$ ms). Curves compare our estimate, ARGUS, the analytic target, and the three controls; coincident curves are omitted where indicated in the panels.

Scope. These probes validate the gravitational and impact terms that the estimator is designed to recover, not every source of contact load. In particular, the estimate omits body-normal internal cloth forces and can therefore underread tension-supported or gripping contact. We evaluate it on the proposal state because the estimated load and the resulting Coulomb correction must refer to the same predicted step; on the impact probe, a realized-state stencil shifts the correct response one step later. The validated quantity is therefore a current-step kinematic estimate rather than an exact contact reaction.

S3. Evaluation protocol

Population. The matched suite crosses four HOOD garment templates [4]—t-shirt, tank top, long-sleeve top, and dress—with AMASS CMU motions [7] {01_01, 07_02, 104_54, 05_06, 05_08, 136_12, 133_02, 79_04, 127_02, 143_27} at $\mu \in \{0, 0.1, 0.3, 0.6\}$. Five motions come from the SNUG/HOOD evaluation lineage; five were internally pre-specified from body-only orientation and kinematic criteria before any garment simulation. The term *pre-registered* in Sec. 5.1 refers to this internal ordering, not a public registry: the metric definitions preceded coefficient-dependent measurements for this generation of runs, and the extension rule preceded its garment rollouts.

The reported Cartesian grid retains only motions for which PPF completes all four coefficients on all four garments. PPF failures were deterministic contact walls or linear-conditioning aborts; two lower-body garments were omitted because too few motions completed. Selection depends only on reference completion, not on neural rollouts or metric values. Every method uses the motion file's native SMPL shape coefficients.

Contact metrics. One judge marks a vertex as contacting when its point–triangle signed body distance is at most 3 mm—the operator's proximity shell ε of Table S1, adopted as the metric's contact margin—for both neural candidates and PPF. Each contacting vertex-frame contributes its lumped mass times the frame length: the stick fraction is the share at or below ϵ_v, and S is the corresponding mean tangential cloth–body speed over all 40 sequences. Because each method produces its own contact samples, $S(\mu)/S(0)$ uses each arm's own frictionless rollout to isolate its coefficient response; Fig. S2 also reports the absolute levels.

Solver and material conventions. Neural rollouts use $h = 1/30$ s. PPF uses a 10 ms step, at least 32 Newton iterations, and a 2 mm contact gap; it is treated as a fixed-regime physics reference rather than ground truth. For material matching, the HOOD membrane gives areal small-strain coefficients $\mu_A = 11.092$ and $\lambda_A = 20.868\,\mathrm{N\,m^{-1}}$ after applying its 4.7×10^{-4} m thickness. Matching the small-strain limit of PPF's ARAP shear term, $\psi = \frac{1}{2}\mu\sum_i(s_i - 1)^2$ over the principal stretches, to these coefficients doubles the shear entry, and PPF reads stiffness normalized by the shared areal density $\rho_A = 0.20022\,\mathrm{kg\,m^{-2}}$; the matched coefficients are $\bar{\mu} = 2\mu_A/\rho_A$ and $\bar{\lambda} = \lambda_A/\rho_A$. Inverting the solver's isotropic parametrization $\bar{\mu} = E/(2(1+\nu))$, $\bar{\lambda} = E\nu/\big((1+\nu)(1-2\nu)\big)$ then gives

$$\begin{aligned} \nu &= \frac{\lambda_A}{2\,(\lambda_A + 2\mu_A)} = 0.24236, \\ E &= 2\bar{\mu}\,(1+\nu) = 275.302, \end{aligned} \tag{S3}$$

where E is the density-normalized membrane stiffness PPF consumes, in $\mathrm{m^2\,s^{-2}}$, and ν is the argument of that parametrization rather than a measured fabric Poisson ratio. Equating the two dihedral bending energies likewise gives PPF's bending value $k_b/(4\,s_b\,\rho_A) = 3.8649$ from HOOD's bending coefficient $k_b = 3.962 \times 10^{-5}$ and PPF's fixed bending scale $s_b = 1.28 \times 10^{-5}$. The mapping matches the criterion's cantilever displacement to 0.05 % of strip length (RMS) and its closed-form axial response to a ratio of 0.98.

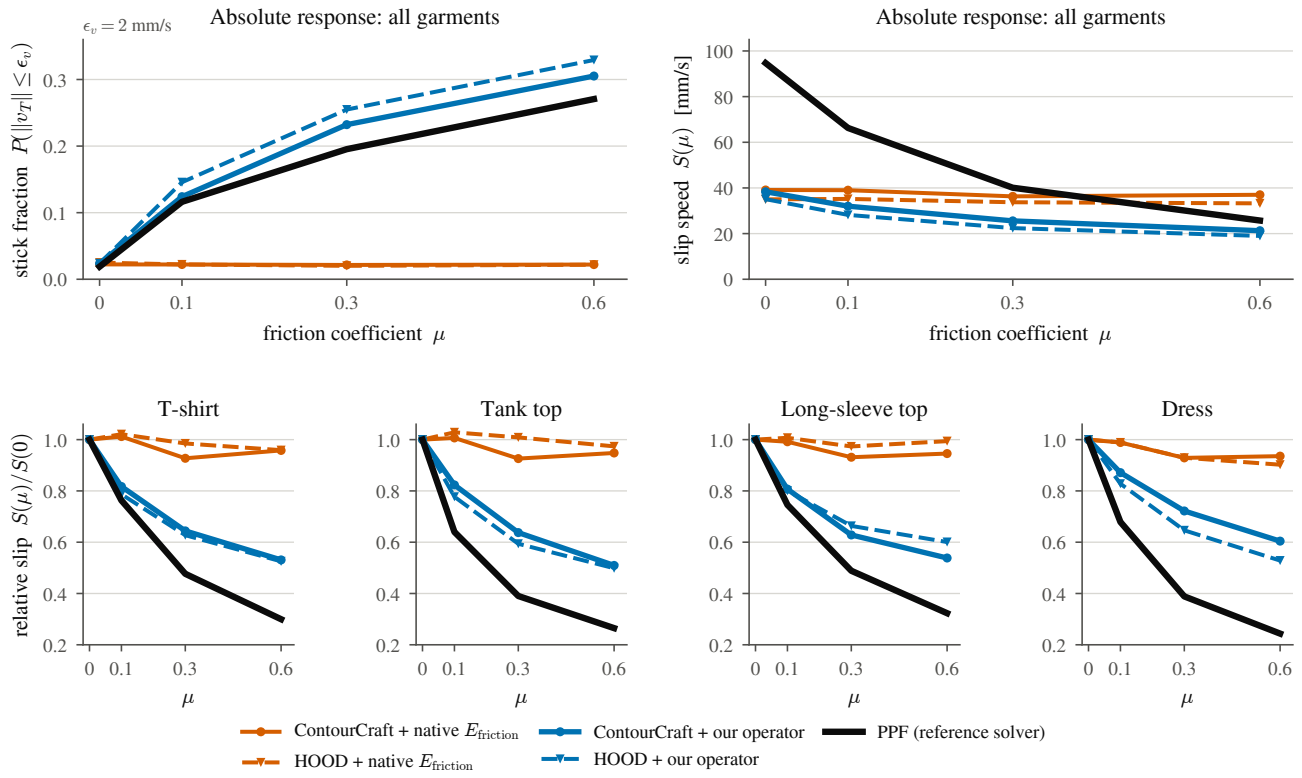


Figure S2. **Absolute and per-garment frictional response.** The top panels pool absolute stick fraction and slip speed over all 40 sequences; the bottom panels pool each garment's ten motions and normalize slip by that arm's own $\mu = 0$ rollout. All panels use the mass–time estimator of Fig. 5 and include every body-orientation bin. Native E_{friction} points are coefficient-specific checkpoints; each operator curve uses one frozen backbone with μ supplied at inference.

S4. Additional quantitative results

At $\mu = 0$, pooled slip is 94.7 mm/s for PPF, 38.3 mm/s for ContourCraft with our operator, and 35.1 mm/s for HOOD with our operator, which motivates the within-arm normalization in the main figure. Figure S2 shows the operator's monotone response on both backbones and every garment, while native retraining remains nearly flat. The operator remains shallower than PPF, and the dress and long-sleeve responses show that the remaining gap depends on both garment and backbone.

S4.1. Sensitivity to the added-strain limit

We compare $\gamma_{\max} \in \{0.1, 0.2, 0.5\}$ at $\mu = 0.6$ over the complete 40-sequence grid, holding the checkpoint, contact set, initialization, and evaluation settings fixed. The primary measure is each sequence's deviation from PPF in stretching-energy ratio. Tightening the limit from 0.2 to 0.1 lowers the mean deviation from 17.3 % to 17.1 % and the added-folding proxy from 14.7 % to 9.0 %; loosening it to 0.5 raises the respective means to 20.0 % and 25.6 %. The lower values at 0.1 reflect stronger clipping, but do not by themselves make it a better operating point: 0.1 lies below the calibrated 15–25 % strain scale of propagated Coulomb corrections in Fig. S3(a) and would truncate the intended correction more aggressively. The limiter does not change the predictor's original deformation, because it acts only on the added field $\mathbf{w}$. We therefore retain the independently calibrated, pre-evaluation value 0.2, which lies at the ordinary correction scale and within the stable calibration region.

S5. Tested learned alternatives

The main comparison uses coefficient-specific checkpoints trained with each backbone's native E_{friction}. We also test one coefficient-conditioned network with the estimated load and IPC's C^1 friction mollifier [6],

$$D_\mu(\hat{\mathbf{x}}^{t+1}) = \sum_{i \in \mathcal{C}_\varepsilon} \mu \, \mathrm{sg}[\widehat{\lambda}_{n,i}] f_0\big(\|\boldsymbol{\Delta}_{T,i}(\hat{\mathbf{x}}^{t+1})\|\big)\,, \quad \text{(S4)}$$

where sg holds the load fixed during differentiation. This route supplies μ to one network without an inference-time operator; unlike a solver, it does not repeatedly update state,

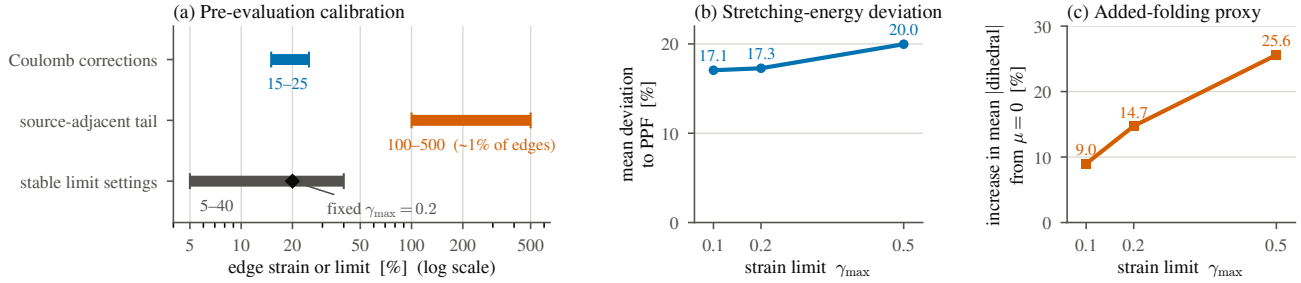


Figure S3. **Calibration and sensitivity of the added-strain limit.** **(a)** Pre-evaluation ranges: edge strain induced by the propagated Coulomb corrections, the source-adjacent tail (roughly 1 % of edges), and limit settings with stable calibration outcomes; the diamond marks the fixed $\gamma_{\max} = 0.2$. **(b,c)** Full-grid means at $\mu = 0.6$ over 40 sequences for deviation from PPF in stretching-energy ratio and for the increase from $\mu = 0$ in mean absolute interior-edge dihedral angle, used as an added-folding proxy. Lower is better for these deformation guardrails, but a tighter limit can improve both simply by suppressing more of the intended friction-induced motion.

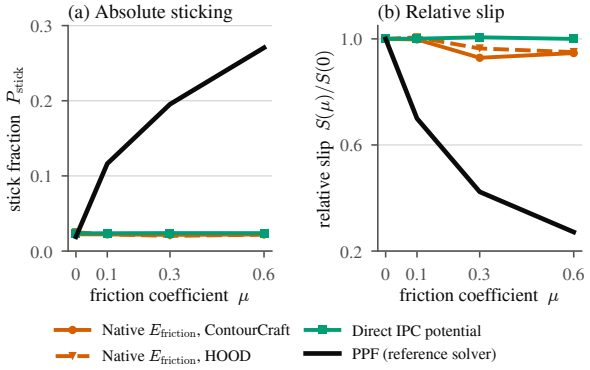


Figure S4. **Loss-only response** on the matched suite, pooled over contact mass and time. Native curves use coefficient-specific checkpoints; direct IPC uses one conditioned checkpoint. Slip is normalized by each candidate's $\mu = 0$ rollout; PPF gives the reference scale.

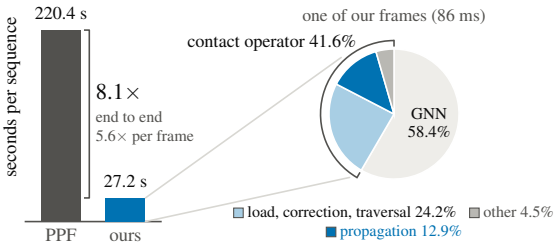


Figure S5. **Runtime at $\mu = 0.6$.** Left: per-sequence wall clock; the end-to-end comparison includes PPF's settling prologue, while the marginal comparison does not. Right: one method frame partitioned into the GNN forward pass; local load estimation with correction and traversal; nonlocal propagation; and the *other* share—contact-pair search with bookkeeping. Frame-weighted means over 12 randomly sampled sequences.

active set, tangent frame, or reaction.

Figure S4 shows that the direct potential remains flat: pooled stick fraction spans 0.0004, and its slip ratio at $\mu = 0.6$ is 1.000. One sequence diverged at $\mu = 0.3$; excluding it leaves this response unchanged, so we omit its energy aggregate. We also trained through the explicit operator and retained it at inference, adding the operator's Coulomb dissipation to the training objective with weight α. At $\mu = 0$, where the operator is inert, training changes pooled slip from the 39.0 mm/s of the three arms' shared seed checkpoint to 21.7, 22.3, and 29.4 mm/s for $\alpha \in \{0.01, 0.1, 1\}$, showing that training already perturbs the frictionless regime. Together with the direct potential's flat coefficient response, this result identifies distinct limitations in the two tested formulations; neither establishes that friction cannot be learned.

S6. Runtime details

At $\mu = 0.6$, the operator's busiest evaluated coefficient, the frame-weighted mean over 3,168 frames from 12 randomly sampled sequences is 86.39 ms, of which the GNN forward pass takes 50.45 ms and the operator 35.93 ms. Timings are wall-clock with device synchronization at stage boundaries, on one NVIDIA RTX 4090 (24 GB) with an Intel Xeon w5-2455X CPU and 128 GB RAM. Of the operator time, local load estimation, correction, and traversal take 20.92 ms; nonlocal propagation takes 11.14 ms; the remaining 3.87 ms—the *other* share in Fig. S5—is contact-pair search and bookkeeping outside the stages themselves.

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